

Supplementary Material for “Dual-Scale Co-evolutionary Algorithm for Multimodal Multiobjective Optimization”

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This supplementary material contains Sections S1–S7, cited from the main text as “the supplementary material”. Tables and figures carry section-based labels (Tables S1a–S1c, S3, S4a, S4b, S5, S5b, S7a, S7b; Figures S2-1–S2-20, S5-1, S5-2, S7-1, S7-2). References are collected at the end of the document.

S1. PER-INSTANCE COMPLETE RESULTS

This section collects the per-instance results of the main comparison. Tables S1a and S1b report the per-instance medians for the two metrics, and Table S1c lists the full problem suite.

Notes to Tables S1a and S1b:

- **Scope.** All 71 benchmark problems in six groups (MMF / IDMP / Local-front / Polygon / Mode-count / HDMMF) $\times$ eight algorithms: DS-MMEA and seven external baselines. Two metrics are reported: IGD_X (decision-space IGD, lower is better) in Table S1a and IGD (objective-space IGD, lower is better) in Table S1b. Both tables have the same 71 rows, one row per problem.
- **Value in each cell.** Median over 10 independent runs (runs 1–10) of that algorithm on that problem, shown with 4 significant digits.
- **Run window.** DS-MMEA carries 20 runs on the nine Polygon problems; the other seven algorithms have 10 runs each. To keep every cell of this table on one identical protocol, all columns uniformly use runs 1–10; the reinforcing runs 11–20 are not used here.
- **Pairing.** Run k of every algorithm uses the identical seed ($20260907 + 1000k + \text{problem index}$), so the significance tests below are strictly paired by this seed table.
- **Symbols.** Each baseline cell shows its median followed by (+), (−) or ($\approx$): the outcome of a two-sided Wilcoxon signed-rank test on the 10 paired per-run values of that algorithm against DS-MMEA. (+) = that algorithm significantly better than DS-MMEA (its median is lower), (−) = significantly worse (its median is higher), ($\approx$) = not separable at $p < 0.05$. DS-MMEA is the reference column and carries no symbol.
- **DNF.** DNF = the algorithm produced no completed run on that problem (deterministic crash; e.g. MMOEADC on NMMF5, whose objective values overflow to Inf on random points). A DNF cell counts as the worst value in any rank-based summary and no significance test applies to it. Problem rows affected by a DNF are marked with a dagger.
- **Reproducibility.** Generated from the archived per-run results by the export scripts of the reproduction package (Section S6); no new experiments. The statistics follow the package’s report script (medians, signed-rank tests, DNF handling).

TABLE S1a: Per-instance IGDX medians (decision-space IGD, lower is better) on the 71 benchmark problems: DS-MMEA versus the seven external baselines. Each cell is the median over the 10 paired runs; a baseline cell carries the outcome of the two-sided Wilcoxon signed-rank test on the 10 paired per-run values against DS-MMEA: (+) the baseline is significantly better, (−) significantly worse, (≈) not separable at $p < 0.05$. DNF marks an algorithm that produced no completed run on that problem (no test applies); † flags the affected problem rows. See the notes at the beginning of Section S1.

Problem (group)	DS-MMEA	HREA	MMOEA	CMMO	TriMOEA-TA&R	HHC-MMEA	HDMMODE	FPITSEA
MMF1 (MMF)	0.04055	0.03897 (+)	0.04529 (−)	0.03968 (≈)	0.06615 (−)	0.07863 (−)	0.06375 (−)	0.04855 (−)
MMF2 (MMF)	0.02533	0.03676 (≈)	0.01806 (+)	0.02509 (≈)	0.05749 (−)	0.2086 (−)	0.01994 (≈)	0.04795 (−)
MMF3 (MMF)	0.03008	0.0313 (≈)	0.01598 (+)	0.02495 (≈)	0.04339 (−)	0.1379 (−)	0.02428 (≈)	0.03452 (−)
MMF4 (MMF)	0.01932	0.02004 (≈)	0.02738 (−)	0.01979 (≈)	0.03838 (−)	0.05753 (−)	0.0384 (−)	0.02391 (−)
MMF5 (MMF)	0.07314	0.06385 (+)	0.07944 (−)	0.07024 (+)	0.1085 (−)	0.1332 (−)	0.1037 (−)	0.08678 (−)
MMF6 (MMF)	0.05953	0.05535 (+)	0.0669 (−)	0.05705 (−)	0.09861 (−)	0.1123 (−)	0.0952 (−)	0.0661 (−)
MMF7 (MMF)	0.0216	0.01997 (≈)	0.02827 (−)	0.02001 (+)	0.0424 (−)	0.04061 (−)	0.03664 (−)	0.01474 (+)
MMF8 (MMF)	0.06076	0.04695 (+)	0.05963 (≈)	0.0546 (≈)	0.3616 (−)	0.3175 (−)	0.0953 (−)	0.1138 (−)
MMF9 (MMF)	0.004501	0.009056 (−)	0.006703 (−)	0.007103 (−)	0.003108 (+)	0.01045 (−)	0.01366 (−)	0.001238 (+)
MMF14 (MMF)	0.0366	0.03945 (−)	0.05237 (−)	0.03989 (−)	0.03633 (≈)	0.0763 (−)	0.05252 (−)	0.0125 (+)
MMF1_e (MMF)	0.4887	0.3391 (+)	0.5232 (≈)	0.312 (+)	1.448 (−)	1.946 (−)	0.378 (≈)	0.7432 (−)
MMF1_z (MMF)	0.02985	0.02655 (+)	0.03131 (≈)	0.02818 (+)	0.08065 (−)	0.07486 (−)	0.04579 (−)	0.04225 (−)
MMF14_a (MMF)	0.04858	0.0476 (+)	0.07778 (−)	0.05108 (−)	0.056 (−)	0.0897 (−)	0.0654 (−)	0.04692 (≈)
IDMPM2T1 (IDMP)	0.0005896	0.0009032 (−)	0.0009172 (−)	0.00111 (−)	0.6733 (−)	0.6732 (−)	0.004571 (−)	0.0263 (−)
IDMPM2T2 (IDMP)	0.000838	0.0008379 (≈)	0.001063 (−)	0.0009324 (−)	0.6733 (−)	0.3618 (−)	0.007846 (−)	0.02533 (−)
IDMPM2T3 (IDMP)	0.001137	0.002962 (−)	0.00197 (−)	0.001765 (−)	0.03413 (−)	0.3627 (−)	0.002587 (−)	0.005515 (−)
IDMPM2T4 (IDMP)	0.0005594	0.001711 (−)	0.0007376 (−)	0.001059 (−)	0.6733 (−)	0.6732 (−)	0.002493 (−)	0.002474 (−)
IDMPM3T1 (IDMP)	0.007246	0.00674 (+)	0.008102 (−)	0.009066 (−)	0.6763 (−)	0.8864 (−)	0.01476 (−)	0.2703 (−)
IDMPM3T2 (IDMP)	0.00704	0.006746 (≈)	0.008157 (≈)	0.008365 (≈)	0.6315 (−)	0.5793 (−)	0.01237 (−)	0.01513 (−)
IDMPM3T3 (IDMP)	0.007546	0.008597 (≈)	0.01032 (≈)	0.01042 (≈)	0.2776 (−)	0.6315 (−)	0.01908 (−)	0.02962 (−)
IDMPM3T4 (IDMP)	0.02346	0.007735 (+)	0.007941 (+)	0.008714 (+)	0.9512 (−)	0.6525 (−)	0.01911 (≈)	0.1425 (≈)
IDMPM4T1 (IDMP)	0.02596	0.0307 (≈)	0.0132 (≈)	0.2786 (≈)	1.19 (−)	1.187 (−)	0.03776 (−)	1.186 (−)
IDMPM4T2 (IDMP)	0.145	0.5655 (≈)	0.01318 (≈)	0.01643 (+)	1.084 (−)	0.974 (−)	0.03762 (≈)	0.9884 (−)
IDMPM4T3 (IDMP)	0.02028	0.2889 (−)	0.01617 (≈)	0.02329 (≈)	0.8697 (−)	0.9729 (−)	0.03604 (≈)	0.7199 (−)
IDMPM4T4 (IDMP)	0.2879	0.4567 (≈)	0.01645 (+)	0.02497 (+)	1.188 (−)	0.9741 (−)	0.04142 (+)	0.8777 (−)
IDMPM2T1_e (Local-front)	0.0005933	0.0006569 (−)	0.0009266 (−)	0.6732 (−)	0.6732 (−)	0.6731 (−)	0.0008552 (−)	0.6732 (−)
IDMPM2T2_e (Local-front)	0.0008568	0.000931 (−)	0.0009589 (−)	0.6731 (−)	0.6732 (−)	0.6733 (−)	0.001171 (−)	0.6732 (−)
IDMPM2T3_e (Local-front)	0.001082	0.001378 (−)	0.001378 (−)	0.3007 (−)	0.3212 (−)	0.4712 (−)	0.003285 (−)	0.3008 (−)
IDMPM2T4_e (Local-front)	0.002077	0.002943 (≈)	0.07352 (−)	1.008 (−)	1.008 (−)	0.5802 (−)	0.2172 (−)	0.7999 (−)
IDMPM3T1_e (Local-front)	0.007422	0.006955 (+)	0.00814 (−)	0.6263 (−)	0.7697 (−)	0.6457 (−)	0.2542 (−)	0.6265 (−)
IDMPM3T2_e (Local-front)	0.008206	0.2535 (−)	0.00789 (≈)	0.4962 (−)	0.8882 (−)	0.8868 (−)	0.13 (−)	0.505 (−)
IDMPM3T3_e (Local-front)	0.008429	0.2561 (−)	0.009027 (≈)	0.4994 (−)	0.769 (−)	1.016 (−)	0.02131 (−)	0.5197 (−)
IDMPM3T4_e (Local-front)	0.0132	0.5041 (−)	0.01145 (≈)	0.8478 (−)	1.011 (−)	0.9392 (−)	0.03912 (≈)	0.8517 (−)
MMF11 (Local-front)	0.006852	0.01147 (−)	0.00766 (≈)	0.2504 (−)	0.2524 (−)	0.2499 (−)	0.009438 (−)	0.2491 (−)
MMF12 (Local-front)	0.003332	0.002761 (+)	0.003079 (≈)	0.2454 (−)	0.2481 (−)	0.2455 (−)	0.003709 (−)	0.2453 (−)
MMF13 (Local-front)	0.08928	0.05221 (+)	0.07853 (≈)	0.2544 (−)	0.2712 (−)	0.2792 (−)	0.05606 (≈)	0.2481 (−)
MMF15 (Local-front)	0.03853	0.04482 (−)	0.05347 (−)	0.2565 (−)	0.2709 (−)	0.2355 (−)	0.05326 (−)	0.2483 (−)
MMF15_a (Local-front)	0.06812	0.05459 (+)	0.08985 (−)	0.2088 (−)	0.2218 (−)	0.2171 (−)	0.1747 (−)	0.2045 (−)
MMF16_l1 (Local-front)	0.05207	0.04883 (≈)	0.06757 (≈)	0.1496 (−)	0.15 (−)	0.1714 (−)	0.06068 (≈)	0.132 (−)
MMF16_l2 (Local-front)	0.1105	0.05812 (+)	0.1142 (≈)	0.3319 (−)	0.3467 (−)	0.3233 (−)	0.1216 (≈)	0.3258 (−)
MMF16_l3 (Local-front)	0.07747	0.05809 (+)	0.09711 (−)	0.206 (−)	0.207 (−)	0.2167 (−)	0.1928 (−)	0.1915 (−)
Polygon_M3_D4 (Polygon)	0.17	3.185 (−)	0.4163 (≈)	0.2616 (≈)	3.387 (−)	4.366 (−)	0.4147 (≈)	0.2784 (≈)
Polygon_M3_D8 (Polygon)	7.556	7.604 (≈)	1.113 (+)	0.7176 (+)	7.775 (−)	8.166 (−)	0.8075 (+)	7.519 (≈)
Polygon_M3_D10 (Polygon)	8.534	8.813 (−)	1.443 (+)	0.877 (+)	8.717 (−)	8.727 (−)	1.682 (+)	8.452 (≈)
Polygon_M4_D4 (Polygon)	0.197	3.336 (−)	0.4759 (−)	0.2976 (−)	2.608 (−)	4.161 (−)	0.4013 (−)	0.3154 (−)
Polygon_M4_D8 (Polygon)	7.415	7.47 (≈)	1.187 (+)	0.7382 (−)	7.602 (−)	7.749 (−)	0.9992 (+)	6.32 (≈)
Polygon_M4_D10 (Polygon)	8.365	8.46 (−)	1.636 (+)	0.9615 (+)	8.709 (−)	9.54 (−)	1.321 (+)	8.45 (≈)
Polygon_M6_D4 (Polygon)	0.2046	3.267 (−)	0.4884 (−)	0.3355 (−)	4.446 (−)	4.666 (−)	0.4249 (−)	0.3782 (−)

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TABLE S1a — continued from the previous page

Problem (group)	DS-MMEA	HREA	MMOEADC	CMMO	TriMOEA-TA&R	HHC-MMEA	HDMMODE	FPITSEA
Polygon_M6_D8 (Polygon)	0.6267	7.532 (−)	1.32 (≈)	0.8267 (≈)	7.683 (−)	8.425 (−)	1.026 (≈)	7.331 (−)
Polygon_M6_D10 (Polygon)	8.06	8.497 (−)	1.67 (+)	1.08 (+)	8.615 (−)	9.588 (−)	1.246 (+)	8.362 (≈)
IDMPT1_NP2 (Mode-count)	0.0005974	0.0007589 (−)	0.0008961 (−)	0.0009609 (−)	0.6732 (−)	0.6733 (−)	0.001375 (−)	0.3588 (−)
IDMPT1_NP4 (Mode-count)	0.001171	0.006366 (−)	0.001905 (−)	0.003141 (−)	0.4239 (−)	0.3273 (−)	0.003894 (−)	0.1589 (−)
IDMPT1_NP8 (Mode-count)	0.03032	0.02708 (≈)	0.003984 (+)	0.007131 (+)	0.1672 (−)	0.2061 (−)	0.00744 (+)	0.02678 (≈)
IDMPT2_NP2 (Mode-count)	0.0008483	0.0009098 (≈)	0.0009932 (−)	0.0008878 (≈)	0.6732 (−)	0.6733 (−)	0.003959 (−)	0.005028 (−)
IDMPT2_NP4 (Mode-count)	0.00125	0.001523 (≈)	0.002097 (≈)	0.001761 (≈)	0.2002 (−)	0.1598 (−)	0.004867 (≈)	0.01181 (−)
IDMPT2_NP8 (Mode-count)	0.01666	0.01521 (≈)	0.004069 (+)	0.004093 (≈)	0.1242 (−)	0.1302 (−)	0.008769 (+)	0.03169 (≈)
NMMF1 (HDMMF)	8.851	8.925 (≈)	17.47 (−)	17.68 (−)	20.1 (−)	14.56 (−)	12.31 (−)	24.61 (−)
NMMF2 (HDMMF)	1.503	2.236 (≈)	0.5707 (+)	0.2529 (+)	4.62 (−)	4.617 (−)	0.07699 (+)	5.39 (−)
NMMF3 (HDMMF)	3.495	5.652 (−)	7.249 (−)	6.105 (−)	6.302 (−)	3.711 (≈)	1.889 (+)	10.23 (−)
NMMF4 (HDMMF)	5.247	23.39 (−)	19.12 (−)	17.22 (−)	5.652 (≈)	2.402 (+)	3.616 (+)	23.38 (−)
NMMF5† (HDMMF)	31.7	36.9 (≈)	DNF	32.16 (≈)	31.68 (≈)	22.8 (+)	22.63 (+)	34.97 (≈)
NMMF6 (HDMMF)	26.61	23.49 (≈)	21.85 (+)	18.48 (+)	10.55 (+)	9.461 (+)	12.31 (+)	27.73 (≈)
NMMF7 (HDMMF)	22.4	34.21 (≈)	21.46 (≈)	17.19 (≈)	21 (≈)	16.13 (+)	11.49 (+)	23.73 (≈)
NMMF8 (HDMMF)	6.96	11.47 (−)	10.24 (−)	7.863 (≈)	10.63 (−)	6.94 (≈)	5.328 (+)	10.96 (−)
NMMF9 (HDMMF)	2.481	3.313 (≈)	2.682 (≈)	2.986 (≈)	9.991 (−)	2.738 (≈)	3.413 (≈)	6.07 (≈)
NMMF10 (HDMMF)	6.609	13.24 (−)	10.23 (−)	10.05 (−)	9.079 (−)	5.094 (+)	1.667 (≈)	16.16 (−)
NMMF11 (HDMMF)	23.58	27.54 (−)	22.32 (≈)	24.19 (≈)	16.7 (≈)	11.78 (+)	13.88 (+)	23.87 (≈)
NMMF12 (HDMMF)	13.51	16.09 (−)	15.1 (−)	15.86 (−)	13.5 (≈)	11.42 (≈)	9.959 (+)	20.88 (−)
NMMF13 (HDMMF)	25.81	40.42 (−)	40.11 (−)	24.45 (+)	23.83 (+)	23.08 (+)	18.92 (+)	49.31 (−)
NMMF14 (HDMMF)	9.721	13.05 (−)	15.41 (−)	11.03 (≈)	14.35 (−)	19.34 (−)	6.268 (+)	19.86 (−)
NMMF15 (HDMMF)	23.91	27.59 (−)	27.01 (≈)	25.26 (≈)	21.91 (≈)	12.8 (+)	18.58 (+)	30.27 (−)

TABLE S1b: Per-instance IGD medians (objective-space IGD, lower is better) on the 71 benchmark problems: DS-MMEA versus the seven external baselines. Each cell is the median over the 10 paired runs; a baseline cell carries the outcome of the two-sided Wilcoxon signed-rank test on the 10 paired per-run values against DS-MMEA: (+) the baseline is significantly better, (−) significantly worse, (≈) not separable at $p < 0.05$. DNF marks an algorithm that produced no completed run on that problem (no test applies); † flags the affected problem rows. See the notes at the beginning of Section S1.

Problem (group)	DS-MMEA	HREA	MMOEA	CMMO	TriMOEA-TA&R	HHC-MMEA	HDMODE	FPITSEA
MMF1 (MMF)	0.0034	0.003581 (≈)	0.003433 (≈)	0.00302 (+)	0.004906 (−)	0.00355 (≈)	0.005833 (−)	0.002418 (+)
MMF2 (MMF)	0.009316	0.0147 (−)	0.009779 (≈)	0.01174 (−)	0.02053 (−)	0.0701 (−)	0.01456 (−)	0.01683 (−)
MMF3 (MMF)	0.007389	0.0125 (−)	0.008542 (≈)	0.01042 (≈)	0.01714 (−)	0.05758 (−)	0.01261 (−)	0.01362 (−)
MMF4 (MMF)	0.003188	0.003318 (≈)	0.00338 (≈)	0.003227 (≈)	0.003435 (≈)	0.0029 (≈)	0.004923 (−)	0.001491 (+)
MMF5 (MMF)	0.00315	0.003592 (−)	0.003592 (−)	0.003207 (≈)	0.004097 (−)	0.003393 (≈)	0.005958 (−)	0.002315 (+)
MMF6 (MMF)	0.003399	0.003524 (≈)	0.003461 (≈)	0.003252 (≈)	0.003853 (−)	0.003063 (+)	0.005803 (−)	0.001836 (+)
MMF7 (MMF)	0.003451	0.003647 (≈)	0.003736 (≈)	0.003057 (+)	0.003913 (−)	0.002684 (+)	0.005593 (−)	0.00102 (+)
MMF8 (MMF)	0.004776	0.003919 (+)	0.002609 (+)	0.003523 (+)	0.003422 (+)	0.002658 (+)	0.005871 (≈)	0.001052 (+)
MMF9 (MMF)	0.01585	0.02806 (−)	0.01052 (+)	0.01229 (+)	0.07166 (−)	0.009595 (+)	0.02592 (−)	0.001531 (+)
MMF14 (MMF)	0.06573	0.06713 (≈)	0.06707 (≈)	0.06248 (+)	0.08653 (−)	0.1046 (−)	0.07988 (−)	0.0163 (+)
MMF1_e (MMF)	0.004211	0.007627 (−)	0.005439 (≈)	0.009685 (−)	0.007019 (≈)	0.01622 (−)	0.009594 (−)	0.005263 (≈)
MMF1_z (MMF)	0.003105	0.003248 (≈)	0.003364 (−)	0.00296 (≈)	0.004266 (−)	0.002821 (+)	0.005574 (−)	0.001532 (+)
MMF14_a (MMF)	0.06961	0.0691 (≈)	0.06593 (+)	0.06559 (+)	0.07853 (−)	0.08965 (−)	0.0812 (−)	0.0386 (+)
IDMPM2T1 (IDMP)	0.0005671	0.0005541 (≈)	0.0009895 (−)	0.000738 (−)	0.0006389 (−)	0.0005491 (≈)	0.001296 (−)	0.0002912 (+)
IDMPM2T2 (IDMP)	0.0005725	0.0005394 (≈)	0.0005674 (≈)	0.0005213 (+)	0.0006564 (−)	0.0004772 (+)	0.001219 (−)	0.0001473 (+)
IDMPM2T3 (IDMP)	0.0006182	0.0005414 (+)	0.0008473 (−)	0.000602 (≈)	0.0006416 (−)	0.0004704 (+)	0.001304 (−)	0.0001555 (+)
IDMPM2T4 (IDMP)	0.0005077	0.0004633 (≈)	0.0006546 (−)	0.0005688 (≈)	0.0006447 (−)	0.0004808 (≈)	0.000948 (−)	0.0002208 (+)
IDMPM3T1 (IDMP)	0.00538	0.005105 (≈)	0.006383 (−)	0.005821 (≈)	0.009727 (−)	0.004993 (+)	0.006218 (−)	0.002479 (+)
IDMPM3T2 (IDMP)	0.004834	0.004508 (+)	0.0055 (−)	0.004778 (≈)	0.009435 (−)	0.004948 (≈)	0.006289 (−)	0.002334 (+)
IDMPM3T3 (IDMP)	0.004843	0.004666 (+)	0.006331 (−)	0.005131 (−)	0.00865 (−)	0.004842 (≈)	0.01181 (−)	0.002281 (+)
IDMPM3T4 (IDMP)	0.004345	0.004644 (≈)	0.005915 (−)	0.00525 (−)	0.009687 (−)	0.00505 (≈)	0.01042 (−)	0.002648 (+)
IDMPM4T1 (IDMP)	0.006278	0.008851 (−)	0.01864 (−)	0.01769 (−)	0.02084 (−)	0.007662 (−)	0.01192 (−)	0.004208 (+)
IDMPM4T2 (IDMP)	0.005477	0.005596 (≈)	0.01527 (−)	0.01848 (−)	0.02036 (−)	0.006628 (−)	0.01912 (−)	0.003597 (+)
IDMPM4T3 (IDMP)	0.005688	0.006834 (−)	0.01584 (−)	0.02117 (−)	0.01875 (−)	0.006709 (−)	0.0427 (−)	0.003852 (+)
IDMPM4T4 (IDMP)	0.005257	0.00595 (−)	0.01951 (−)	0.03149 (−)	0.01998 (−)	0.006907 (−)	0.04494 (−)	0.004575 (≈)
IDMPM2T1_e (Local-front)	0.00086	0.001098 (−)	0.001349 (−)	0.007102 (−)	0.007353 (−)	0.007157 (−)	0.001341 (−)	0.007038 (−)
IDMPM2T2_e (Local-front)	0.0008833	0.0009258 (≈)	0.0008986 (≈)	0.007166 (−)	0.007403 (−)	0.007216 (−)	0.001233 (−)	0.007069 (−)
IDMPM2T3_e (Local-front)	0.001004	0.001146 (−)	0.001075 (−)	0.00498 (−)	0.005141 (−)	0.004979 (−)	0.001644 (−)	0.004788 (−)
IDMPM2T4_e (Local-front)	0.001524	0.001509 (≈)	0.00133 (≈)	0.01592 (−)	0.01625 (−)	0.006337 (−)	0.005 (−)	0.008571 (−)
IDMPM3T1_e (Local-front)	0.006927	0.006785 (+)	0.00758 (−)	0.02408 (−)	0.02806 (−)	0.02502 (−)	0.007871 (−)	0.02389 (−)
IDMPM3T2_e (Local-front)	0.007282	0.01575 (−)	0.008127 (−)	0.03619 (−)	0.04031 (−)	0.03759 (−)	0.0148 (−)	0.03625 (−)
IDMPM3T3_e (Local-front)	0.007726	0.01571 (−)	0.008803 (≈)	0.0362 (−)	0.04017 (−)	0.0376 (−)	0.01628 (−)	0.03616 (−)
IDMPM3T4_e (Local-front)	0.008967	0.01586 (−)	0.00919 (≈)	0.0361 (−)	0.03972 (−)	0.0371 (−)	0.04181 (−)	0.03612 (−)
MMF11 (Local-front)	0.03008	0.0396 (≈)	0.02111 (+)	0.09351 (−)	0.1665 (−)	0.09391 (−)	0.02808 (≈)	0.08906 (−)
MMF12 (Local-front)	0.006287	0.006692 (≈)	0.003896 (+)	0.08324 (−)	0.08547 (−)	0.08321 (−)	0.005541 (≈)	0.08227 (−)
MMF13 (Local-front)	0.0343	0.03247 (≈)	0.02442 (+)	0.1471 (−)	0.2438 (−)	0.1463 (−)	0.02784 (≈)	0.1421 (−)
MMF15 (Local-front)	0.09288	0.1017 (−)	0.1002 (−)	0.1781 (−)	0.2061 (−)	0.2003 (−)	0.1156 (−)	0.1579 (−)
MMF15_a (Local-front)	0.1285	0.1154 (+)	0.1323 (≈)	0.1755 (−)	0.1961 (−)	0.1853 (−)	0.1636 (−)	0.1666 (−)
MMF16_l1 (Local-front)	0.1039	0.09954 (≈)	0.0983 (≈)	0.1487 (−)	0.1723 (−)	0.1723 (−)	0.1108 (≈)	0.1179 (≈)
MMF16_l2 (Local-front)	0.1587	0.132 (+)	0.1317 (+)	0.2326 (−)	0.2614 (−)	0.2474 (−)	0.1584 (≈)	0.2205 (−)
MMF16_l3 (Local-front)	0.1335	0.126 (≈)	0.1256 (≈)	0.1915 (−)	0.2173 (−)	0.2065 (−)	0.185 (−)	0.1696 (−)
Polygon_M3_D4 (Polygon)	0.07789	0.1196 (−)	0.1721 (−)	0.09689 (−)	0.1454 (−)	0.2166 (−)	0.1441 (−)	0.06314 (+)
Polygon_M3_D8 (Polygon)	0.2049	0.2704 (−)	0.5369 (−)	0.2497 (−)	0.275 (−)	0.5948 (−)	0.299 (−)	0.1606 (+)
Polygon_M3_D10 (Polygon)	0.2615	0.3354 (−)	0.7279 (−)	0.3004 (−)	0.3247 (−)	0.4647 (−)	0.4351 (−)	0.2434 (≈)
Polygon_M4_D4 (Polygon)	0.1141	0.1723 (−)	0.2463 (−)	0.1373 (−)	0.2818 (−)	0.2478 (−)	0.1895 (−)	0.08902 (+)
Polygon_M4_D8 (Polygon)	0.2997	0.3632 (−)	0.6911 (−)	0.3024 (≈)	0.405 (−)	0.4992 (−)	0.4081 (−)	0.2325 (≈)
Polygon_M4_D10 (Polygon)	0.3858	0.4865 (−)	1.012 (−)	0.4222 (−)	0.4738 (−)	1.251 (−)	0.5111 (−)	0.2967 (+)
Polygon_M6_D4 (Polygon)	0.158	0.238 (−)	0.3084 (−)	0.1927 (−)	0.3887 (−)	0.343 (−)	0.2546 (−)	0.1284 (+)

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TABLE S1b — continued from the previous page

Problem (group)	DS-MMEA	HREA	MMOEADC	CMMO	TriMOEA-TA&R	HHC-MMEA	HDMMODE	FPITSEA
Polygon_M6_D8 (Polygon)	0.2292	0.5089 (−)	0.9928 (−)	0.4482 (−)	0.6338 (−)	1.447 (−)	0.5053 (−)	0.3336 (−)
Polygon_M6_D10 (Polygon)	0.2959	0.6311 (−)	1.311 (−)	0.6198 (−)	0.7187 (−)	1.785 (−)	0.6514 (−)	0.4071 (−)
IDMPT1_NP2 (Mode-count)	0.0005705	0.0005832 (≈)	0.0009377 (−)	0.0006957 (−)	0.0007412 (−)	0.0005388 (≈)	0.001244 (−)	0.0002614 (+)
IDMPT1_NP4 (Mode-count)	0.0006423	0.0006964 (−)	0.001121 (−)	0.0009237 (−)	0.0006985 (−)	0.0005552 (+)	0.00152 (−)	0.0004393 (+)
IDMPT1_NP8 (Mode-count)	0.000604	0.0007169 (−)	0.001262 (−)	0.0007603 (−)	0.0007012 (−)	0.0005927 (≈)	0.001493 (−)	0.0004617 (+)
IDMPT2_NP2 (Mode-count)	0.000581	0.0005362 (+)	0.0005968 (≈)	0.0005327 (≈)	0.0006886 (−)	0.0004496 (+)	0.001148 (−)	0.0001165 (+)
IDMPT2_NP4 (Mode-count)	0.000605	0.0005778 (≈)	0.0007835 (−)	0.0005813 (+)	0.0006951 (−)	0.000473 (+)	0.001263 (−)	0.0002196 (+)
IDMPT2_NP8 (Mode-count)	0.0005894	0.0006307 (≈)	0.0007529 (−)	0.0006221 (≈)	0.000642 (≈)	0.0005094 (+)	0.00128 (−)	0.000209 (+)
NMMF1 (HDMMF)	5.1	8.373 (−)	5.174 (≈)	4.339 (≈)	6.744 (≈)	3.743 (≈)	8.918 (−)	3.761 (+)
NMMF2 (HDMMF)	160.1	2.155 (+)	0.398 (+)	0.5253 (+)	0.08289 (+)	34.88 (+)	0.04129 (+)	0.1519 (+)
NMMF3 (HDMMF)	0.1411	0.01892 (+)	0.04055 (+)	0.01789 (+)	0.006981 (+)	0.005552 (+)	0.009447 (+)	0.001413 (+)
NMMF4 (HDMMF)	381.5	101.9 (+)	21.44 (+)	15.57 (+)	59.56 (+)	48.37 (+)	0.08995 (+)	52.4 (+)
NMMF5† (HDMMF)	5.132×10^9	1.144×10^{12} (−)	DNF	2.054×10^8 (+)	1.091×10^{11} (≈)	1.077×10^9 (≈)	8.099×10^5 (+)	4.646×10^9 (≈)
NMMF6 (HDMMF)	228.8	95.19 (≈)	18.39 (+)	1.315 (+)	0.3593 (≈)	0.3088 (+)	1.694 (+)	231.7 (≈)
NMMF7 (HDMMF)	286.4	9.93×10^{14} (−)	121.6 (≈)	0.08071 (≈)	23.39 (+)	1.212×10^{10} (≈)	0.4484 (+)	0.09768 (+)
NMMF8 (HDMMF)	1.843	1.997 (≈)	3.455 (−)	2.854 (−)	1.045 (≈)	0.6564 (+)	1.675 (≈)	3.107 (−)
NMMF9 (HDMMF)	2.165	1.574 (≈)	0.3427 (+)	0.4675 (+)	0.4651 (+)	0.727 (+)	0.4464 (+)	0.9073 (+)
NMMF10 (HDMMF)	939.9	4775 (≈)	404.4 (≈)	439.4 (+)	272.3 (+)	400.2 (+)	0.194 (+)	6467 (−)
NMMF11 (HDMMF)	1724	6023 (≈)	639.8 (≈)	370.2 (≈)	20.12 (+)	575.8 (≈)	2047 (≈)	1964 (≈)
NMMF12 (HDMMF)	117.7	300.6 (−)	100.1 (≈)	85.64 (≈)	79.48 (+)	157.2 (≈)	83.94 (+)	287.7 (−)
NMMF13 (HDMMF)	1.233×10^5	1.934×10^5 (≈)	8.216×10^5 (−)	1.26×10^4 (+)	1.033×10^4 (+)	1.683×10^5 (≈)	1.632×10^4 (+)	1.134×10^5 (≈)
NMMF14 (HDMMF)	3.366	322.7 (−)	0.1954 (+)	0.5024 (+)	1.257 (≈)	4.428 (≈)	0.02831 (+)	0.9823 (≈)
NMMF15 (HDMMF)	9.238	11.64 (−)	7.59 (+)	8.149 (+)	8.166 (+)	11.03 (≈)	10.17 (≈)	8.523 (≈)

TABLE S1c: The 71 benchmark problems in six groups: protocol parameters (M objectives, D decision variables, population size N , evaluation budget maxFE), the number of equivalent (global) Pareto sets (Modes), and a one-line characterization. * marks the suites whose smallest box-normalized inter-mode distance lies below the 0.05 resolution of the recovery indicators (all nine Polygon problems and NMMF3); the mode count is exact, but recovery-type indicators degenerate there. Notes follow the table.

Problem	Group	M	D	N	maxFE	Modes	Characterization
MMF1	MMF	2	2	200	10000	1	Sinusoidal PS valley with two symmetric branches; concave front.
MMF2	MMF	2	2	200	10000	2	Two modes split by x_2 ; concave front.
MMF3	MMF	2	2	200	10000	2	Two asymmetric modes; concave front.
MMF4	MMF	2	2	200	10000	2	Two modes split by $\sin(\pi - x_1)$; convex front.
MMF5	MMF	2	2	200	10000	2	Two modes on a $\sin(6\pi)$ valley; concave front.
MMF6	MMF	2	2	200	10000	1	PS branches over alternating basins in x_1 ; concave front.
MMF7	MMF	2	2	200	10000	1	Wavy-ridge PS with two symmetric branches; concave front.
MMF8	MMF	2	2	200	10000	2	Two modes; circular front ($f_1 = \sin - x_1$).
MMF9	MMF	2	2	200	10000	2	Two modes via $\sin(2\pi x_2)^6$; hyperbolic front.
MMF14	MMF	3	3	300	15000	2	Three objectives; spherical front; two modes in the last variable.
MMF1_e	MMF	2	2	200	10000	1	MMF1 (two PS branches) with the x_2 range widened to $[-20, 20]$.
MMF1_z	MMF	2	2	200	10000	1	MMF1 (two PS branches) with the x_2 range narrowed to $[-1, 1]$.
MMF14_a	MMF	3	3	300	15000	2	MMF14 variant defined on the unit box.
IDMPM2T1	IDMP	2	2	200	10000	2	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM2T2	IDMP	2	2	200	10000	2	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM2T3	IDMP	2	2	200	10000	2	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM2T4	IDMP	2	2	200	10000	2	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM3T1	IDMP	3	3	300	15000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM3T2	IDMP	3	3	300	15000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM3T3	IDMP	3	3	300	15000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM3T4	IDMP	3	3	300	15000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM4T1	IDMP	4	4	400	20000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM4T2	IDMP	4	4	400	20000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM4T3	IDMP	4	4	400	20000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM4T4	IDMP	4	4	400	20000	4	IDMP: equivalent PSs at unequal distances from the center; search-difficulty imbalance.
IDMPM2T1_e	Local-front	2	2	200	10000	2	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM2T2_e	Local-front	2	2	200	10000	2	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM2T3_e	Local-front	2	2	200	10000	3	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM2T4_e	Local-front	2	2	200	10000	7	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM3T1_e	Local-front	3	3	300	15000	4	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM3T2_e	Local-front	3	3	300	15000	4	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM3T3_e	Local-front	3	3	300	15000	4	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
IDMPM3T4_e	Local-front	3	3	300	15000	8	IDMP with additional local PSs (local fronts); difficulty imbalance kept.
MMF11	Local-front	2	2	200	10000	2	Two modes plus local Pareto fronts (Gaussian-envelope multimodality).
MMF12	Local-front	2	2	200	10000	8	Two modes plus local Pareto fronts (Gaussian-envelope multimodality).
MMF13	Local-front	2	3	300	15000	2	Two objectives; two global modes plus local fronts (Gaussian $\times$ $\sin^6$ landscape).
MMF15	Local-front	3	3	300	15000	2	Three objectives; spherical front; two global modes plus local fronts.
MMF15_a	Local-front	3	3	300	15000	2	Three objectives; spherical front; two global modes plus local fronts.

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TABLE S1c — continued from the previous page

Problem	Group	M	D	N	maxFE	Modes	Characterization
MMF16_l1	Local-front	3	3	300	15000	3	Three objectives; two global PSs plus local PSs (local-front variants).
MMF16_l2	Local-front	3	3	300	15000	3	Three objectives; two global PSs plus local PSs (local-front variants).
MMF16_l3	Local-front	3	3	300	15000	4	Three objectives; two global PSs plus local PSs (local-front variants).
Polygon_M3_D4	Polygon	3	4	200	20000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M3_D8	Polygon	3	8	200	40000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M3_D10	Polygon	3	10	200	50000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M4_D4	Polygon	4	4	200	20000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M4_D8	Polygon	4	8	200	40000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M4_D10	Polygon	4	10	200	50000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M6_D4	Polygon	6	4	200	20000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M6_D8	Polygon	6	8	200	40000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
Polygon_M6_D10	Polygon	6	10	200	50000	4*	Polygon: four equivalent PSs from an M-vertex mapping; high-dimensional, many-objective.
IDMPT1_NP2	Mode-count	2	2	200	10000	2	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
IDMPT1_NP4	Mode-count	2	2	200	10000	4	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
IDMPT1_NP8	Mode-count	2	2	200	10000	8	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
IDMPT2_NP2	Mode-count	2	2	200	10000	2	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
IDMPT2_NP4	Mode-count	2	2	200	10000	4	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
IDMPT2_NP8	Mode-count	2	2	200	10000	8	IDMP mode-count sweep: NP equivalent PSs (NP = 2/4/8) on one problem template.
NMMF1	HDMMF	2	7	200	35000	12	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF2	HDMMF	2	10	200	50000	2	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF3	HDMMF	2	10	200	50000	21*	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF4	HDMMF	2	10	200	50000	1	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF5	HDMMF	2	16	200	80000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF6	HDMMF	2	12	200	60000	2	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF7	HDMMF	2	16	200	80000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF8	HDMMF	2	12	200	60000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF9	HDMMF	2	25	200	125000	3	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF10	HDMMF	2	16	200	80000	2	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF11	HDMMF	2	18	200	90000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF12	HDMMF	2	18	200	90000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF13	HDMMF	3	60	200	300000	2	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF14	HDMMF	2	10	200	50000	16	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).
NMMF15	HDMMF	2	30	200	150000	4	HDMMF: high-dimensional multimodal problem (decision dimension 7-60).

Notes to Table S1c:

- 1) **Protocol** (identical for all algorithms, per `rq3_problems.m`): population $N = 100 D$ for MMF / IDMP / Local-front, $N = 200$ for Polygon / Mode-count / HDMMF; budget $\text{maxFE} = 5000 D$ for every group (Mode-count: $10000 = 5000 \times 2$).

- 2) **Modes** = number of equivalent (global) Pareto sets given by the unified ground truth CoSEATruePS ($N = 2000$ reference samples; analytic per-mode generators where the problem provides them, connectivity clustering otherwise). Counts match the expected values hardcoded in `rq3_problems.m` wherever those are specified.
- 3) ***** = the smallest box-normalized distance between two distinct modes is below the 0.05 resolution of the recovery indicators (all 9 Polygon problems, 0.023, and NMMF3, 0.024); the mode count is still exact, but the recovery indicators degenerate on these problems, so IGD_X-type metrics are the meaningful comparison. Connected PSs that double back on themselves (e.g. MMF1, MMF6, MMF7, NMMF4) count as one mode under connectivity clustering although they consist of two symmetric branches.
- 4) Every cell was read from the on-disk truth cache; no problem evaluation was re-run for this table.

S2. FINAL-POPULATION DISTRIBUTION PLATES ON 20 REPRESENTATIVE INSTANCES

A. Scope, layout, and conventions

This section collects the final-population distribution plates for 20 representative instances of the 71-instance suite (Table II of the main text), eight algorithms per instance. The 20 instances were selected by a rule fixed before the plates were drawn: per-group quotas of four instances from the MMF group, three from IDMP, four from local-front, three from Polygon, two from mode-count, and four from HDMMF; within each group the instances are placed evenly along the within-group ranking of DS-MMEA in the main comparison, and the best- and worst-ranked instance of every group is always included. The plates show final populations of single runs, so they complement rather than duplicate the per-instance statistics of Section S1.

Each plate is a 4-row by 4-column panel: two algorithms per row, with the decision-space and objective-space panels of each algorithm side by side. The eight algorithms appear in a fixed reading order (left to right, then top to bottom): DS-MMEA (this work), HREA [1], MMOEADC [2], CMMO [3], TriMOEA-TA&R [4], HHC-MMEA [5], HDMMODE [6], and FPITSEA [7]. Every cell overlays the ground-truth reference set (the same samples used by the scoring metrics), drawn in gray beneath the population. For each algorithm, the displayed run is the one of the 10 paired runs whose IGD_X lies closest to that algorithm's own median on the instance; the run index and the IGD_X value of the displayed run are printed in the title of the decision-space panel, and the IGD of the same run in the title of the objective-space panel.

Projection and plotting conventions:

- 1) Decision space, more than three variables: the reference samples and the eight final populations of the instance are merged, box-normalized, and subjected to one principal-component analysis per instance; every cell of the instance shows the first two principal components under this shared projection, so all sixteen cells of a plate are directly comparable. With three or fewer variables the populations are plotted directly in the original coordinates.
- 2) Objective space, more than three objectives: parallel coordinates over the objective axes. With three or fewer objectives the population is plotted directly.
- 3) The objective-space panels of a plate share one pair of axis ranges across the eight algorithms. Solutions pressed against a panel edge are genuine outliers and are kept there deliberately; the rendering does not crop them away.
- 4) Solutions with non-finite objective values are removed before plotting.
- 5) A run that stored no finite solutions is rendered as a “DNF” (did not finish) tile; the treatment of such runs in the metric tables follows Section S1.

B. Reading caveats

- 1) **Known objective mismatch.** On NMMF6 the objective functions implemented in the platform differ from the reference data distributed with the original proposal [6]. The objective-space column of that plate is indicative only; comparisons on this instance should be read from the decision-space column and from the metric tables of Section S1.
- 2) **Instances where recovery rates degenerate.** On the nine Polygon instances the smallest box-normalized distance between two true modes lies below the 0.05 resolution of the mode-recovery indicator (Section III of the main text), so recovery-type indicators degenerate there and IGD_X-type distances are the meaningful comparison. On the HDMMF instances the 0.05 membership threshold is far below the pairwise distances of a 200-point population in 7–60 dimensions, so membership-based counts degenerate there as well; distance-based readings are used instead (Fig. S7-2).
- 3) **One run per cell.** Each cell shows the median-closest run of that algorithm, not an average population; cross-run variability is quantified by the paired statistics of Section S1 and is not visible in a single plate.

C. Plates

Captions state the group, the number of objectives M , the number of decision variables d , and the number of true modes, per Table II of the main text.

Low-dimensional MMF group (4 instances):

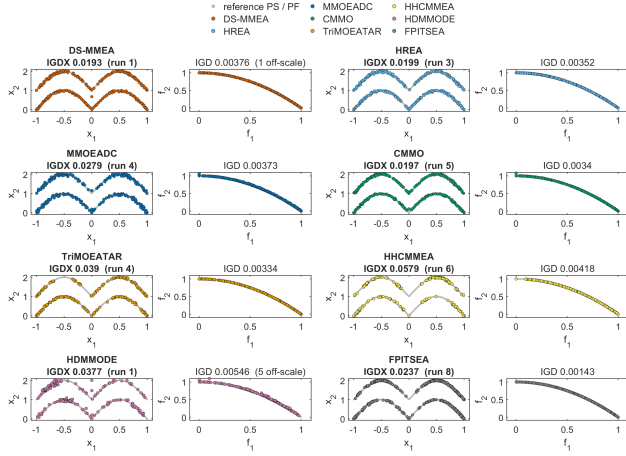


Fig. S2-1: MMF4 — two modes, $M = 2$, $d = 2$.

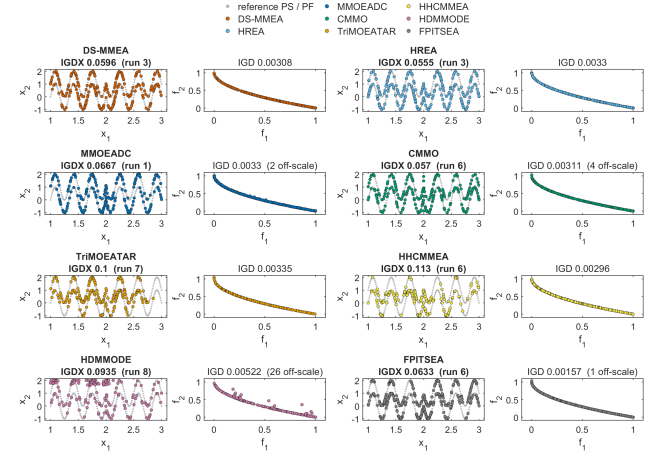


Fig. S2-2: MMF6 — one connected (two-branch) mode, $M = 2$, $d = 2$.

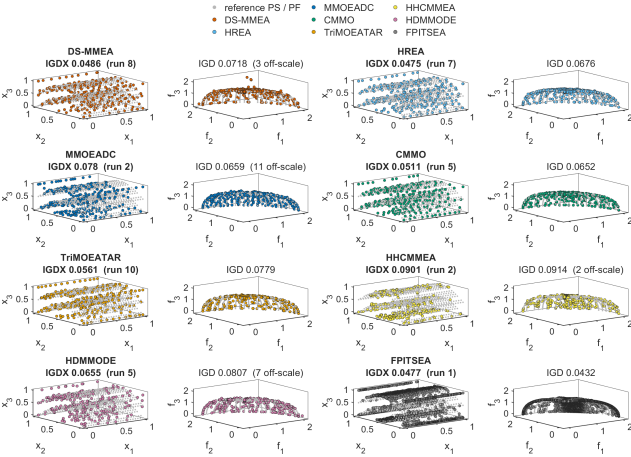


Fig. S2-3: MMF14_a — two modes, three objectives, $M = 3$, $d = 3$.

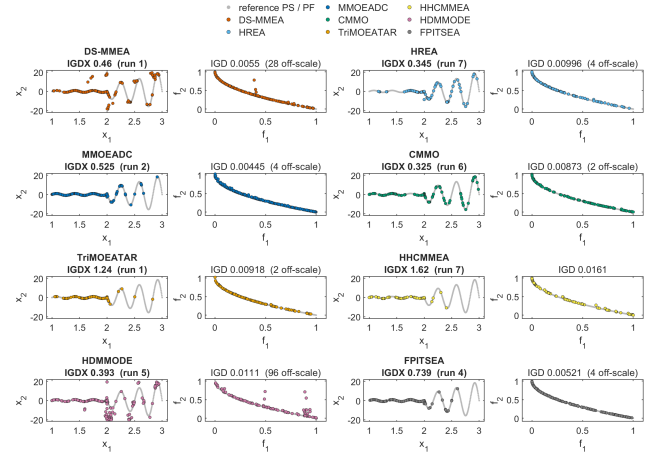
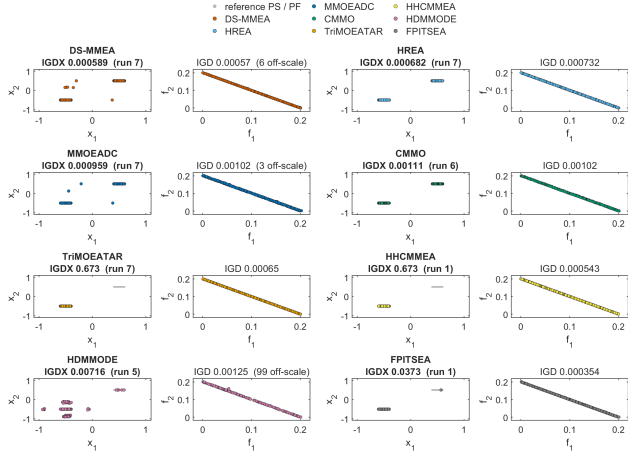
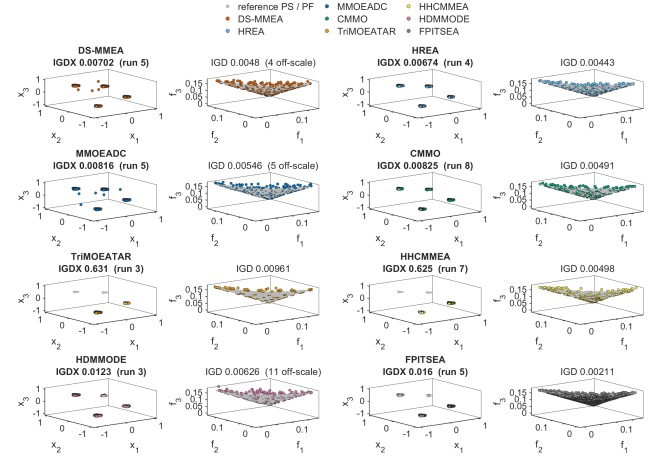
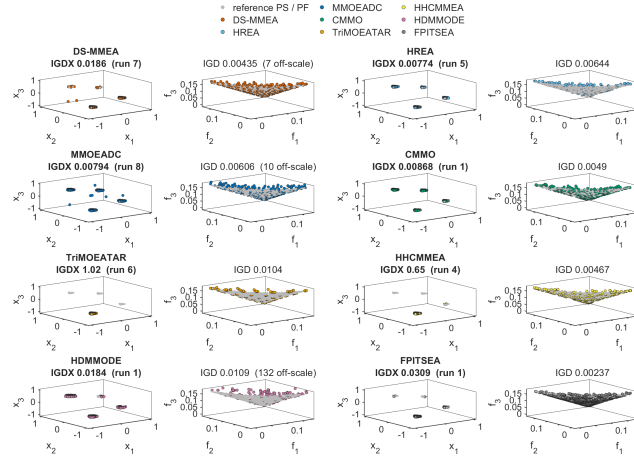


Fig. S2-4: MMF1_e — one connected (two-branch) mode, widened variable range, $M = 2$, $d = 2$.

IDMP group (3 instances):**Fig. S2-5:** IDMPM2T1 — two equivalent Pareto sets at unequal search difficulty, $M = 2$, $d = 2$.**Fig. S2-6:** IDMPM3T2 — four equivalent Pareto sets at unequal search difficulty, $M = 3$, $d = 3$.**Fig. S2-7:** IDMPM3T4 — four equivalent Pareto sets at unequal search difficulty, $M = 3$, $d = 3$.

Local-front group (4 instances):

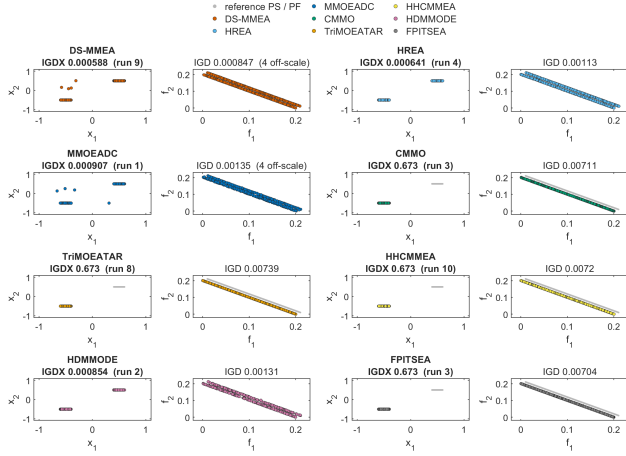


Fig. S2-8: IDMPM2T1_e — IDMP with additional local Pareto sets, $M = 2$, $d = 2$.

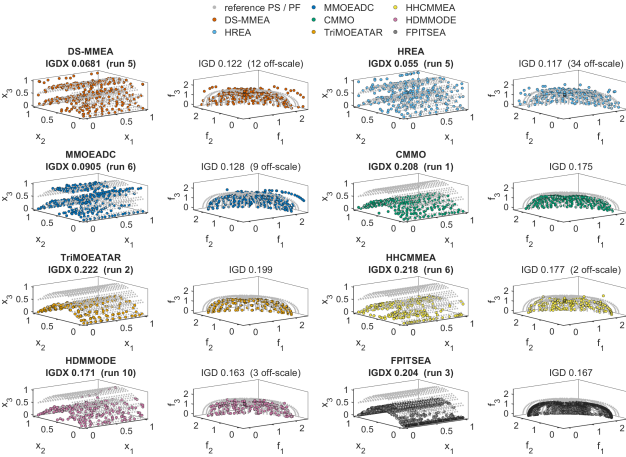


Fig. S2-10: MMF15_a — two global modes plus local fronts, unit-box variant, $M = 3$, $d = 3$.

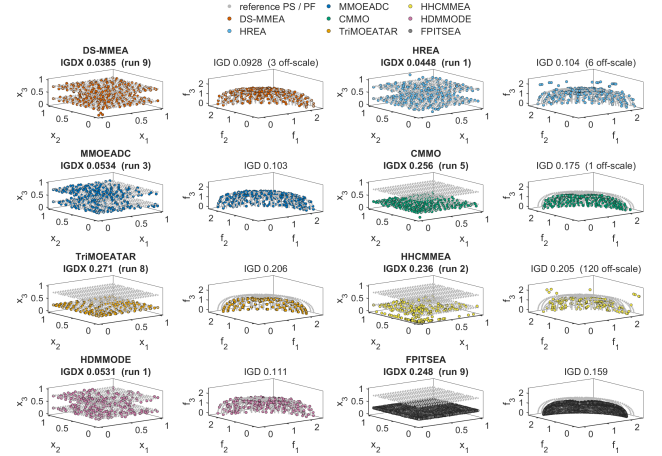


Fig. S2-9: MMF15 — two global modes plus local fronts, $M = 3$, $d = 3$.

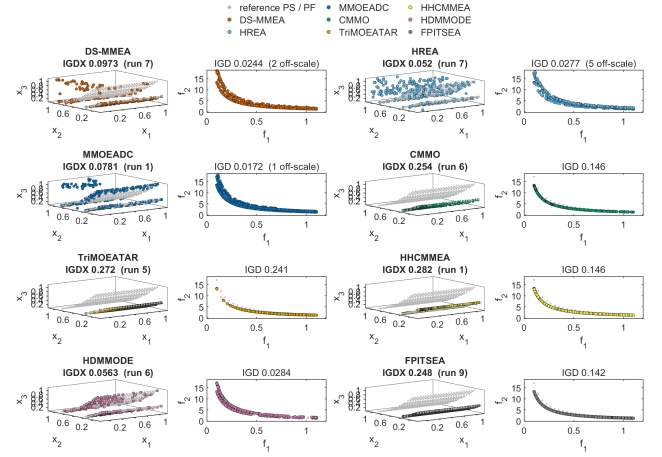


Fig. S2-11: MMF13 — two global modes plus local fronts, $M = 2$, $d = 3$.

Polygon group (3 instances):

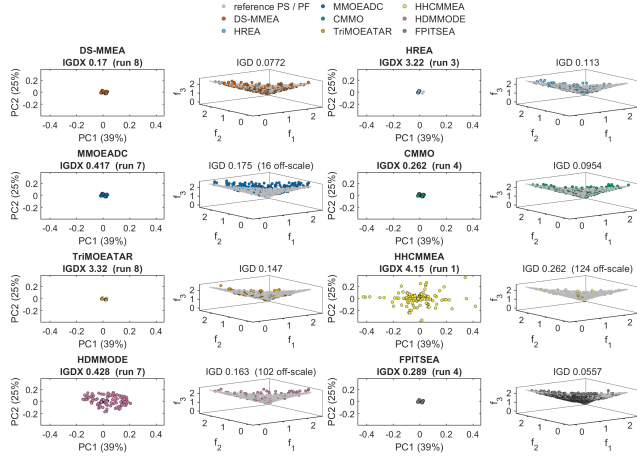


Fig. S2-12: Polygon_M3_D4 — four equivalent Pareto sets, $M = 3$, $d = 4$.

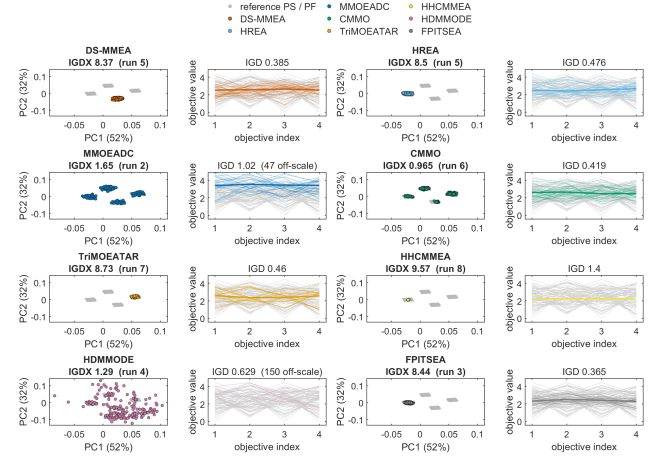


Fig. S2-13: Polygon_M4_D10 — four equivalent Pareto sets, $M = 4$, $d = 10$.

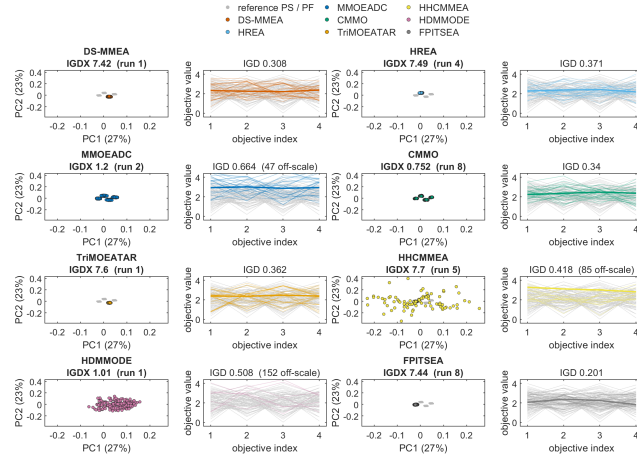


Fig. S2-14: Polygon_M4_D8 — four equivalent Pareto sets, $M = 4$, $d = 8$.

IDMP mode-count sweep (2 instances):

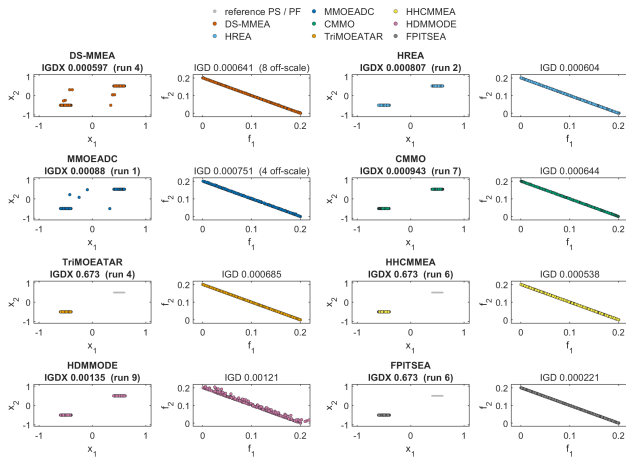


Fig. S2-15: IDMP_T1_NP2 — two equivalent Pareto sets, $M = 2$, $d = 2$.

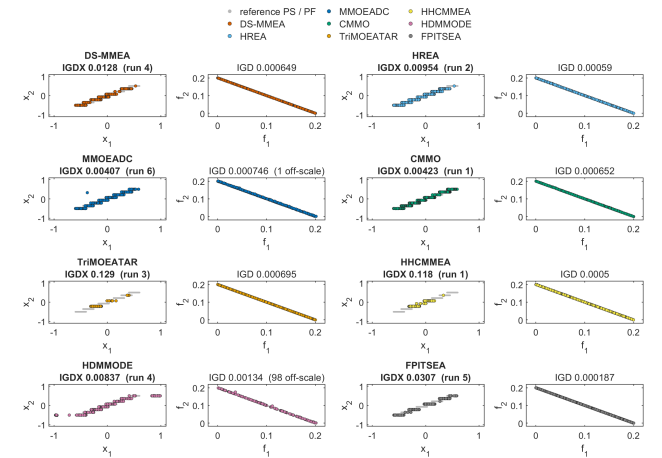
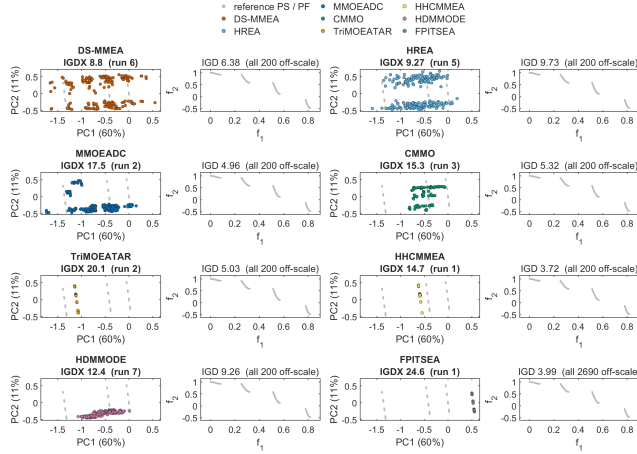
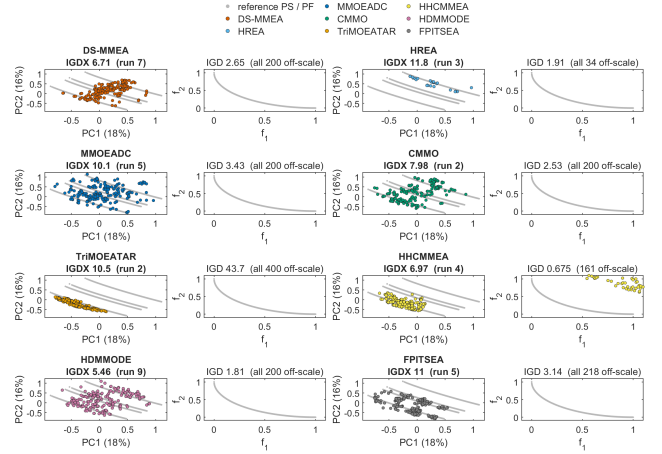
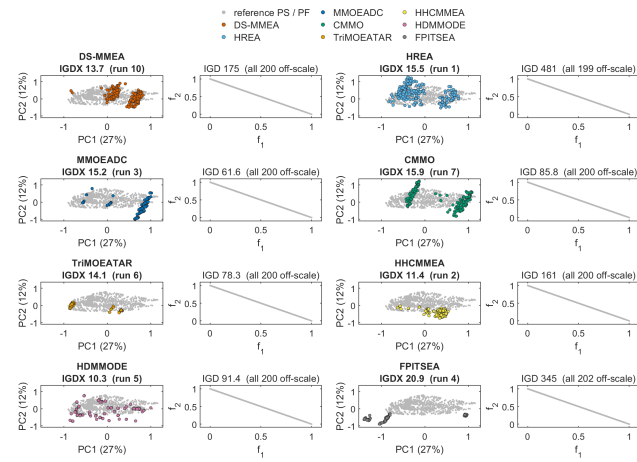
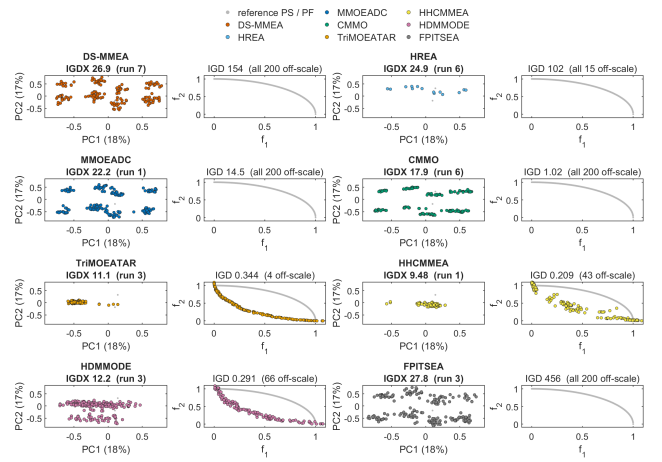


Fig. S2-16: IDMP_T2_NP8 — eight equivalent Pareto sets, $M = 2$, $d = 2$.

HDMMF group (4 instances):**Fig. S2-17:** NMMF1 — twelve modes, $M = 2$, $d = 7$.**Fig. S2-18:** NMMF8 — four modes, $M = 2$, $d = 12$.**Fig. S2-19:** NMMF12 — four modes, $M = 2$, $d = 18$.**Fig. S2-20:** NMMF6 — two modes, $M = 2$, $d = 12$; the objective-mismatch caveat of Section S2.2 applies.**S3. THE STAGED ELIMINATION STUDY OF TWELVE CANDIDATE RULES — SUMMARY RECORD**

All counts and ratios in this section are quoted from the statistics recorded when the study was run; see the data-retention note at the end of this section.

A. Scope and protocol

Section IV of the main text reports the outcome of a staged elimination study that searched for a single clustering rule (and gate form) able to serve all three mode geometries at once: curve-like modes, compact high-dimensional modes, and sparse nascent colonies. Twelve candidate rules were evaluated in two stages against the single-family configuration of the algorithm: nine candidates in stage one (six clustering replacements and three gate replacements, each on a panel of diagnostic instances drawn from the three geometries) and three combination candidates in stage two (a data-adaptive radius cap crossed with three gate forms, on a 24-instance panel). Every cell used ten paired runs over a fixed seed table, roughly 2,300 runs in total.

Ratios are quoted as the median IGDX of the candidate divided by the median IGDX of the reference configuration of the corresponding stage, so values below 1 favor the candidate. In stage one the reference is the single-family configuration (fixed radius with the annealed gate); in stage two the reference is the gate-free single-family variant, which stage one had identified as the stronger baseline on the catastrophe instances. Win/loss tallies such as “9W/1L” (9 of 10 paired runs better, 1 worse) count the paired runs of one diagnostic instance.

B. Summary table of the twelve candidates

Table S3 compiles the record. “Repair face” lists the geometries the candidate was designed to fix and its recorded diagnostic results; “fragmentation face” lists the geometries it damaged.

TABLE S3: The staged elimination study of twelve candidate rules: summary record. Ratios are the median IGD_X of the candidate divided by the median IGD_X of the stage reference (values below 1 favor the candidate); “9W/1L” tallies count the paired runs of one diagnostic instance. “None recorded” marks a recorded verdict without retained cell statistics; see the data-retention note at the end of this section.

#	Candidate	Mechanism (one sentence)	Repair face	Fragmentation face	Verdict
<i>Stage one — clustering replacements (reference: fixed radius + annealed gate)</i>					
1	Union k -nearest-neighbor graph	Connected components of the union k -NN graph, $k = \sqrt{n}$; radius-free.	None recorded.	—	Eliminated in stage one.
2	Median order-statistic radius	Radius = $2 \times$ the median nearest-neighbor distance, recomputed every generation.	Fusion-heavy diagnostic instances repaired, $0.46 \times - 0.64 \times$.	Dense low-dimensional curve instances degraded $1.28 \times$; a continuous Pareto set was cut into 34–48 pseudo-modes and the development channel stalled.	Rejected in stage one.
3	Whitened-PCA median radius	The same order-statistic radius applied in the whitened principal-component space.	None recorded.	—	Eliminated in stage one.
4	Q90 order-statistic radius	Radius = $2 \times$ the 90th percentile of the nearest-neighbor distances.	Safest on low-dimensional curves (MMF1 $0.94 \times$, MMF13 $1.05 \times$); high-dimensional repair partial only (Polygon_M4_D10 $0.99 \times$).	None recorded; the candidate failed by insufficiency, not by harm.	Rejected in stage one (early-run spread inflates the statistic it uses).
5	Mutual k -nearest-neighbor graph	Connected components of the intersection (mutual) k -NN graph, $k = \sqrt{n}$; radius-free, zero new parameters.	Polygon_M4_D10 $0.62 \times$ (9W/1L, recovery rate 0.25 to 1.00), Polygon_M6_D8 $0.40 \times$ (8W/2L), NMMF2 $0.46 \times$ (10W/0L), MMF1 $0.96 \times$.	Fundamental on curve-like modes: MMF16_12 at $0.31 \times$ versus 0.090 for the gate-free variant in the stage-two check — the mutual graph shatters continuous Pareto sets at density necks.	Stage-one winner; rejected in stage two, dismissing the whole graph family.
6	Manifold quota cap	Per-mode development quota capped at $0.1N$ (a fix for the positive feedback by which the largest modes attracted a disproportionate share of the samples).	IDMPM4T1 $0.78 \times$.	Polygon_M6_D8 degraded $2.70 \times$; inert while modes are healthy.	Not adopted; kept as a default-off guard field.
<i>Stage one — gate replacements (reference: annealed global gate)</i>					
7	Within-mode grid ranking	Survivors are re-ranked inside their own mode on an epsilon-grid; no global pruning at all.	IDMPT1_NP4 $0.22 \times$ (0.0996 to 0.0219, recovery rate 1.00 — $2.8 \times$ better than complete gate removal on this instance); NMMF2 $0.50 \times$ (9W/1L).	Local-front diagnostic instances $1.25 \times - 1.41 \times$; MMF1_e $4.13 \times$ (that failure occurred under the fused state of the incumbent clustering).	Stage-one winner among the gates; dominated by complete removal in stage two.
8	Mode-level adjudication	The gate rules on whole modes instead of single members.	Indistinguishable from the incumbent gate on every diagnostic instance.	—	Eliminated in stage one.
9	Outlier-robust normalization	The gate’s normalization range is taken from the global first front itself, immune to objective-space outliers.	Net negative alone: it re-arms the strict global gate and harms weak-colony instances.	Weak-colony regressions.	Rejected as a standalone fix; the normalization repair only matters combined with gate removal.
<i>Stage two — combinations on the capped radius (reference: gate-free single-family variant; 24 instances)</i>					
10	Capped radius + within-mode grid ranking	Radius capped to $\min\{0.08\sqrt{D}, 2Q_{90}\}$, crossed with candidate 7.	Shared record of the capped-radius family, see below.	Shared record of the capped-radius family, see below.	Net negative; rejected.
11	Capped radius + complete gate removal	The same cap crossed with pure mode-relative selection.	As above.	As above.	Net negative; rejected.
12	Capped radius + retained annealed gate	The same cap crossed with the incumbent gate.	As above.	As above.	Net negative; rejected.

The three capped-radius candidates share one recorded repair/fragmentation account. On the repair side: IDMPT1_NP4 $0.03 \times$, IDMPT2_NP8 $0.26 \times$, Polygon_M4_D8 $0.37 \times$, Polygon_M6_D8 $0.40 \times$, and MMF1_e $0.28 \times$ (10W/0L). On the harm side: MMF11 $4.2 \times$, IDMPM3T3_e $4.8 \times$, IDMPM2T1 $4.5 \times$, IDMPM4T3 $2.6 \times$, and MMF8 $1.5 \times - 2.0 \times$. The mechanism is

the one anticipated in Section IV of the main text: on a sparse but continuous mode, large nearest-neighbor distances signal low sampling density rather than a mode boundary, so the cap amputates the mode (seventy-plus pseudo-modes appeared within single runs), and the development channel loses its modeling targets.

C. Outcome

No candidate satisfied all three geometries. Every clustering replacement was net negative: the candidates that repaired one geometry broke another, and the remaining candidates failed by insufficiency. Every gate replacement was dominated by deleting the gate outright. The adopted outcome was therefore deletion, not replacement: the incumbent fixed radius kept, the gate removed entirely, and the three incompatible scale demands carried by two parallel scale hypotheses, which is the architecture of Section V of the main text.

Data-retention note: The raw per-run records of this diagnostic study were not retained once its conclusions had been recorded. Table S3 was compiled from the statistics recorded at the time the study was run; no entry in this table can be recomputed from the archived data of the main comparison. Recorded items without cell statistics are listed with their verdict only (“none recorded”). The archived results of the adopted configuration that this study motivated are, in contrast, fully retained and are the basis of every other table in this paper.

S4. COMPLETE PARAMETER TABLE AND SENSITIVITY NOTES

The fields are defined in `DS-MMEAConfig.m` of the reproduction package (Section S6), which starts from the single-family base configuration and adds ten fields.

A. Overview

DS-MMEA is configured by one structure. It starts from the configuration of the single-family base algorithm, whose inherited fields keep their default values unchanged, and adds ten fields, of which exactly two are active parameters: the number of families (`nPop`) and the clustering rule of the additional family (`clusterModeB`). One further field, `nnQuantileB`, is pinned at its calibrated constant (90) and is discussed below; it is an explicit override that detaches the tight family from an inherited default belonging to a rejected candidate rule, not a free parameter. All remaining new fields carry inert defaults reserved for the ablation switches of Section VII of the main text (the island variant and the three-scale variant) and the sensitivity variants. With `nPop = 1` the main loop degenerates, step for step, to the single-family base; the regression gate of Section S6 pins this property. The configuration exposes no global state and is safe under `parfor`.

Four runtime configurations are used in the paper: the full algorithm; the manifold-off configuration `DS-MMEA-nm` (development channel disabled); the island variant (offspring pool dealt to the families alternately); and the three-scale variant (a third family at a tighter order-statistic radius). Their field settings are marked in the Status column below.

B. The ten new fields

TABLE S4a: New fields added by DS-MMEA. Defaults in brackets are inherited values (empty means “inherit”).

Field	Default	Meaning	Status
<code>nPop</code>	2	Number of parallel scale families. With 1, the run is bit-identical to the single-family counterpart DS-MMEA-1F; with 3, the run is the three-scale variant of Sections S4–S5.	Active
<code>clusterModeB</code>	capped radius	Clustering rule of families 2.. <code>nPop</code> : the fixed radius capped by a data-adaptive order statistic, $\epsilon_t = \min\{0.08\sqrt{D}, 2Q_{90}\}$; the cap only ever tightens. (The exact enum literals of the released source are quoted in the code repository.)	Active
<code>nnQuantileB</code>	90	Order statistic of the nearest-neighbor distances used by the family-2 cap. The inherited default (50, the median) belongs to the rejected median-radius candidate and is explicitly overridden.	Pinned constant (not a parameter)
<code>epsFracB</code>	[]	Family-2 radius base override; [] inherits family 1 (0.08).	Inert (sensitivity variant)
<code>radMultB</code>	[]	Family-2 cap multiplier override; [] inherits family 1 (2.0).	Inert (sensitivity variant)
<code>shareOffspring</code>	true	All families select from one common offspring pool (budget unchanged). false deals offspring to the families alternately: the island variant.	Inert (ablation switch)
<code>clusterModeC</code>	[]	Family-3+ clustering rule; [] inherits family 2.	Inert (three-scale variant)
<code>epsFracC</code>	[]	Family-3+ radius base; [] inherits family 2.	Inert (three-scale variant)
<code>radMultC</code>	[]	Family-3+ cap multiplier; the three-scale variant sets 1.0 (a third scale at the bare Q_{90} radius).	Inert (three-scale variant)

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TABLE S4a — continued from the previous page

Field	Default	Meaning	Status
nnQuantileC	[]	Family-3+ order statistic; [] inherits family 2.	Inert (three-scale variant)

C. Inherited fields (defaults unchanged from the single-family base)

TABLE S4b: Inherited fields, grouped by function. “Inert” means the default leaves the behavior unchanged and the field is consumed only by a non-default variant.

Field	Default	Meaning	Status
<i>Discovery channel: scheduling</i>			
injectEnabled	true	Center-symmetric reflection injection on/off.	Active
kFrac	0.05	Injection cadence: one injection event every round($0.05 \times$ generations) (at least every 2 generations).	Active
injectStartFrac	0.10	Injection window opens at 10% of the budget.	Active
injectStopFrac	1.00	Injection window closes at this budget fraction (1.00 = never closes early).	Active
mFrac	0.30	Upper size of one injection batch: round($0.30 \times N$) candidates.	Active
mMinFrac	0.05	Floor of the adaptive injection batch.	Active
adaptiveInj	true	Survival-rate feedback of the injection batch.	Active
alpha	0.25	Batch growth factor when survival is high.	Active
sHi	0.20	Survival rate above which the batch grows.	Active
sLo	0.02	Survival rate below which the batch halves.	Active
zeroWin	3	Consecutive zero-survival events before the channel retires itself.	Active
<i>Reflection proposal quality</i>			
proposeMode	'reflect'	Proposal operator: center-symmetric reflection. 'random' is the destructive control variant.	Active
randomImmigrant	false	Destructive control: evaluated random points replace reflection candidates.	Inert (control variant)
noiseDecay	true	Residual noise of reflected candidates decays with progress.	Active
sigma0	0.02	Initial noise scale of reflected candidates.	Active
sigmaEnd	0.005	Final noise scale of reflected candidates.	Active
srcWeight	true	Weight the choice of reflection sources by their cluster size.	Active
minClusterQuota	2	Minimum source-cluster size for a reflection source.	Active
<i>Selection backbone</i>			
envSel	SPEA2-style truncation	Diversity truncation rule of the backbone [8].	Active
noveltyFilter	false	Pre-selection novelty filter for injected candidates.	Inert (off by default)
injEpsFrac	0.15	Radius of that filter ($0.15\sqrt{D}$) when enabled.	Inert (off by default)
mateRestrict	false	Pool-restricted mating of the first generation; retired in favor of unrestricted merged mating.	Inert (legacy switch)
epsRetain0 / epsRetainMin / epsRetainCapFrac	0.30 / 0.02 / 0.10	Epsilon-retention truncation schedule; consumed only on the disabled legacy path.	Inert (legacy fields)
<i>Mode identification, family 1</i>			
clusterMode	'eps'	Family-1 clustering rule: connected components under the fixed radius.	Active
epsFrac	0.08	Family-1 radius base: $\epsilon_w = 0.08\sqrt{D}$.	Active
knnK	[]	Neighbor count of the graph clustering variants ($[] = \sqrt{n}$); not consumed by the fixed-radius or capped-radius rules.	Inert under the defaults
matchThrFrac	1.0	Matching radius of the cross-generation mode tracker ($1.0 \times \epsilon\sqrt{d}$).	Active
<i>Gate fields (removed mechanism)</i>			
gateType	'none'	Pure mode-relative selection with no global pruning; the adopted outcome of the elimination study (Section S3). 'global' re-arms the annealed gate as an ablation variant.	Active, fixed by the study
epsAnneal0	0.60	Annealed gate epsilon at progress 0.	Ablation variant ('global')
epsFloor	0.25	Annealed gate epsilon floor (late budget).	Ablation variant ('global')
annealShape	logarithmic schedule	Gate annealing schedule: epsilon anneals as $-\log_2(1.5t)$ with budget fraction t , loose while modes develop and tight late; a linear schedule is kept as the variant's alternative.	Ablation variant ('global')
robustNorm	false	Gate normalization from the global front's own range (the rejected standalone fix of Section S3).	Ablation variant ('global')

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TABLE S4b — continued from the previous page

Field	Default	Meaning	Status
modeFloor	0	Ungated per-mode first-front floor; rejected guard (near-inert, chaos-dominated), kept as an ablation variant.	Ablation variant
modeFloorMaxFrac	0.5	Size bound: the floor applies only to modes at most half of N .	Ablation variant
<i>Development channel</i>			
manifoldEnabled	true	Local-PCA development channel on/off (off = the DS-MMEA-nm configuration).	Active
manifoldFrac	0.35	Per-mode quota share: at most 35% of the mode's size.	Active
minModeSize	8	Maturity threshold: minimum members.	Active
minModeAge	3	Maturity threshold: minimum tracked generations.	Active
varThresh	0.90	Latent dimension covers 90% of the cumulative spectral variance.	Active
residMax	0.20	Residual-variance validity bound of a local model.	Active
maniNoise0 / maniNoiseEnd	1.0 / 0.25	Residual-noise multiplier schedule of sampled candidates.	Active
maniFeedback	true	Per-mode quota survival feedback.	Active
maniSLo / maniSHi	0.02 / 0.20	Hysteresis bands: halve below 0.02, recover by 25% above 0.20, retire at 0.	Active
maniQCapFrac	1.0	Absolute per-mode quota cap as a fraction of N (1.0 = uncapped); the default-off guard field of the elimination study.	Inert (guard field)
<i>Family-1 adaptive radius fields</i>			
radMult	2.0	Cap multiplier of the order-statistic radii; inherited by radMultB. Consumed only under non-default family-1 cluster modes.	Inert under the defaults
nnQuantile	50	Order statistic of the family-1 adaptive radius; inherited default, overridden for family 2.	Inert under the defaults

One further field, `mrEnabled`, is forced to true in DS-MMEA: the algorithm has no global-selection path, and the corresponding ablation lives in the base algorithm's code path.

D. Sensitivity notes

- 1) **The tight family's order statistic is a calibration, not a tuned parameter.** `nnQuantileB` is pinned at 90 because the two candidate statistics separate cleanly: the median (the rejected candidate of Section S3) tracks the body of the nearest-neighbor distribution and shatters dense low-dimensional curve modes, whereas the 90th percentile tracks the spacing tail and leaves curves whole while still shrinking high-dimensional radii by roughly an order of magnitude. The multiplier 2.0 and the quantile 90 were fixed by the stage-two calibration of the elimination study and were not re-tuned in any later phase.
- 2) **The three-scale rejection is a single-point result, read parsimoniously.** The three-scale variant was raced at one parameterization (third radius multiplier 1.0, the tightest natural choice); the space of third-scale settings was not exhausted. The rejection therefore does not claim that no useful third scale exists anywhere; it claims that at the tested point the third scale added wall-clock cost and run-to-run instability without revealing a mode geometry that the two retained scales could not already cover, which is the parsimonious reading of the two-scale design. The counts are in Section S5.

E. Main loop

The main loop of DS-MMEA is given as Algorithm 1 of the main text (Section V-A).

S5. ABLATION DETAILS AND EFFICIENCY BREAKDOWN

Counts from the paired-run archives; timing and budget figures from the solo-timing panel; both tables are part of the reproduction package (Section S6). This section details Table IV of the main text and the efficiency figures of Section VII-D of the main text.

A. Counting conventions

Two counting units appear in this section and must not be mixed. Rows 1 and 2 of Table S5 are per-problem counts over all 71 instances: for every instance, the runs shared by the two configurations (identical seeds) are paired, the per-instance IGD_X medians decide better/worse, and the p-value is a per-instance Wilcoxon signed-rank test on the paired runs, with the median of the 71 p-values reported (the test returns $p = 1$ when all paired differences within an instance are zero). Rows 3 and 4 are per-run counts from the 24-instance subset of the architecture phase: 240 run pairs per configuration, counted run by run. The raw per-run data of this subset were not retained after it concluded, so rows 3 and 4 quote the counts as recorded and cannot be recomputed; they are kept for the contribution record.

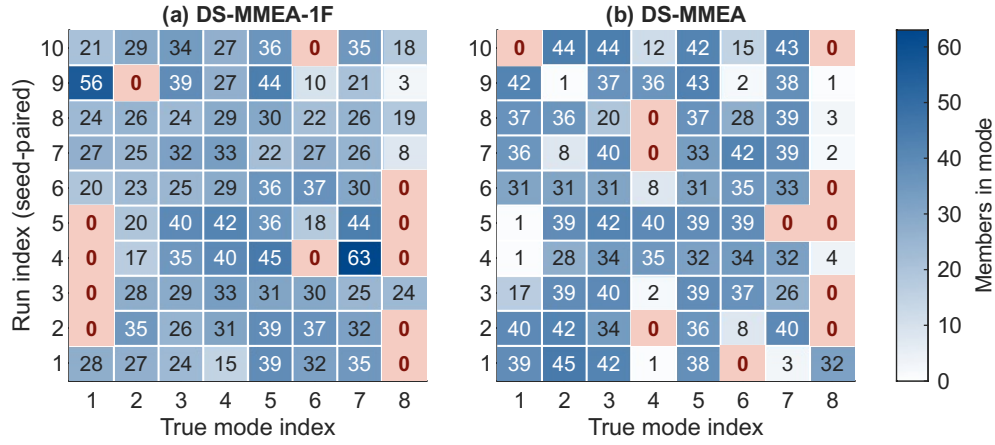


Fig. S5-1: Members of the delivered set in each of the eight true modes of `IDMPT1_NP8` (columns) for each of the 10 seed-paired runs (rows): (a) DS-MMEA-1F, (b) DS-MMEA. Cell color encodes the number of delivered members in the mode (colorbar on the right); cells with zero members are highlighted. DS-MMEA-1F leaves 12 of the 80 mode-run cells empty; DS-MMEA leaves 11 on the delivered basis, and 5 with the output filter disabled (main text, Section VIII).

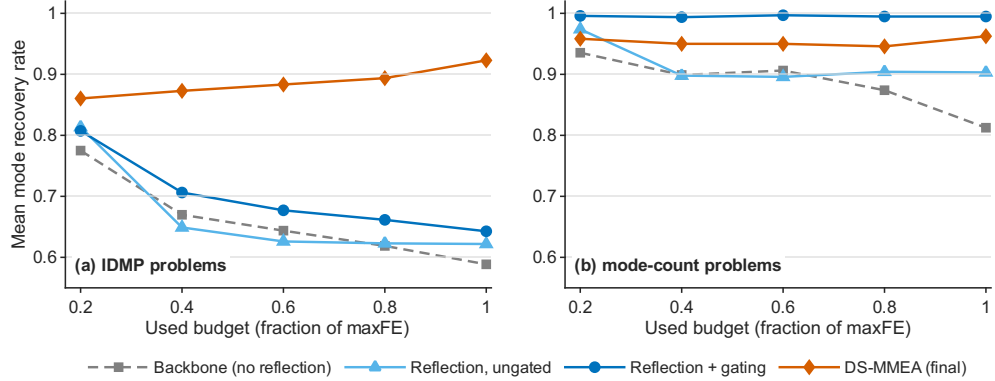


Fig. S5-2: Discovery-channel ablation: mean mode recovery rate versus the consumed fraction of the evaluation budget on the (a) IDMP group (12 problems) and (b) mode-count group (6 problems). Each point averages the 10 independent runs of every problem in the group under the shared seed table (main text, Section VI); this is a group-level mechanism view, whereas all comparisons elsewhere use the per-problem median convention of Section III. The backbone and the two reflection variants are configurations of the single-family base algorithm: the backbone disables reflection entirely, and the two variants add center-symmetric reflection without and with survival-rate feedback, respectively; DS-MMEA (final) is the complete algorithm.

Run counts are 10 per instance throughout, except the nine Polygon instances, which follow the extended protocol of 20 paired runs (Section VI of the main text) wherever both sides carry the 20 runs — in Table S5 this applies to row 1’s Polygon instances.

Naming: DS-MMEA-1F is the single-family counterpart (the family count set to 1; see the regression gates of Section S6); DS-MMEA-nm is the manifold channel disabled; the island variant deals the common offspring pool to the two families alternately (`shareOffspring = false`); the three-scale variant runs a third family at the bare order-statistic radius (Section S4).

B. Ablation counts

TABLE S5: Paired ablations of DS-MMEA (IGDX).

Comparison	Scope	Better	Tie	Worse	Median p	$\#(p < 0.05)$
vs. DS-MMEA-1F (architecture)	71 instances	44	0	27	0.275	24
vs. DS-MMEA-nm (manifold channel)	71 instances	46	0	25	0.193	24
vs. island variant (no shared offspring)	24-instance subset, 240 run pairs	123	–	117	–	–
vs. three-scale variant	24-instance subset, 240 run pairs	124	–	116	–	–

Ties can arise only when two per-instance medians coincide exactly; none occurred in the measured rows. The three-scale row is counted from DS-MMEA’s perspective (the recorded direction “116/124” names the variant’s wins first).

TABLE S5b: Single-run wall-clock time in seconds, measured strictly sequentially (no parallel workers, no contention): 3 runs per cell, 4 representative instances, mean of the 3 runs per cell; the Mean column is the unweighted mean of the 4 cells. Rows marked (a) are ablation or reference configurations outside the 8-algorithm main comparison.

Configuration	MMF1	IDMPM3T2	IDMPM2T2_e	Polygon_M6_D10	Mean
DS-MMEA	2.30	11.49	5.55	16.19	8.88
HREA	4.80	12.55	15.11	32.16	16.16
MMOEADC	2.57	3.21	1.94	9.63	4.34
CMMO	2.41	4.27	2.10	10.35	4.78
TriMOEA-TA&R	0.81	1.60	0.93	3.58	1.73
HHC-MMEA	1.17	1.38	0.92	14.50	4.49
HDMODE	—	—	—	—	—
FPITSEA	—	—	—	—	—
DS-MMEA-1F (a)	1.40	4.27	2.39	8.52	4.15
DS-MMEA-nm (a)	2.25	4.50	2.59	9.75	4.77

C. Efficiency detail

Two configurations carry no entry: no uncontended sequential measurement exists for HDMODE and FPITSEA, and timings taken from contended parallel batches are not comparable. DS-MMEA-nm remains 15% slower than the single-family base (4.77 versus 4.15 s), the residual price of the second environmental selection and of filtering the union of two families rather than one. The full algorithm costs about $2.1\times$ the single-family base (8.88 versus 4.15 s), the price of running and selecting for two families; it remains below the cost of the slowest baselines on this panel.

Budget audit: Over all 71 instances and 10 runs each, DS-MMEA consumes at most $1.020\times$ the nominal evaluation budget (measured maximum 1.0199, on IDMPM2T1 repetition 10), with every candidate injected by the discovery and development channels included in the accounting. This claim is made for DS-MMEA only. Two external baselines exceed the nominal budget through their own internal accounting on the HDMMF group — TriMOEA-TA&R up to $1.667\times$ on NMMF6 and FPITSEA up to $1.641\times$ on NMMF11 — reported as measured; all algorithms started from the same nominal budget.

S6. REPRODUCIBILITY STATEMENT

A. Code availability

The algorithm is implemented in MATLAB on the PlatEMO platform [9] and includes the implementations of all ablation variants reported in this paper. The source code, the per-run archives, and the scripts that generate every table and figure from those archives are publicly available at <https://github.com/Wenhua-Li/DS-MMEA>.

B. Platform, protocol, and seeds

All main experiments ran on one desktop machine with 12 physical cores, using a parallel pool of up to 20 workers; the DS-MMEA-nm configuration of Table IV was re-run on a second machine (a 128-core server, 96 workers) under the identical seed table, and its per-instance verdicts reproduce the original same-machine record exactly (46 versus 25), which bounds cross-context variability; batch jobs store one result file per run and skip runs already on disk, so an interrupted batch resumes without duplicating evaluations. The problem suite (71 instances, 6 groups), the population sizes, and the evaluation budgets are those of Table II of the main text; every algorithm, including all baselines, ran under the same platform protocol with the same budget accounting, and every candidate solution was evaluated through the platform’s evaluation counter (no free evaluations).

Runs are paired through a fixed seed table. The random seed of repetition r on the problem with suite index p is

$$\text{seed} = 20260907 + 1000r + p, \quad (1)$$

with r and p counted from 1. Every configuration compared in one table uses the identical seed table, so all comparisons in this paper are paired by construction: a paired comparison uses only the runs the two configurations share, reports per-instance medians over the shared runs, and tests per instance with a Wilcoxon signed-rank test on the paired runs. Unless stated otherwise, each cell holds 10 repetitions; the nine Polygon instances follow the extended protocol of 20 paired repetitions, adopted after a guard-field probe showed that on such trajectory-chaotic instances a 2–5 member perturbation can flip a 10-run median, so mechanism-level comparisons there require at least 20 paired repetitions.

C. Two-level bit-level regression gates

The implementation is guarded by two bit-level regression gates, both rerun after every code change:

- 1) **Single-family degeneration.** With the family count set to 1, DS-MMEA must reproduce the complete per-generation evaluation sequence of its single-family counterpart (DS-MMEA-1F) bit for bit, on a fixed set of problems and seeds.

TABLE S7a: Manifold channel activity per group (medians over problems and runs, from the per-generation logs of the archived runs).

Group	Active gens.	Offspring share	Early	Middle	Late
MMF	0.90	0.136	0.091	0.173	0.087
IDMP	0.91	0.302	0.215	0.342	0.342
Local-front	0.91	0.283	0.197	0.331	0.330
Polygon	0.85	0.137	0.111	0.150	0.150
Mode-count	0.92	0.254	0.186	0.289	0.293
HDMMF	0.00	0.001	0.003	0.000	0.000

Note. “Active gens.” = share of generations with at least one manifold offspring; “offspring share” = manifold offspring as a fraction of the offspring pool. On the HDMMF group the last generation with activity sits at 0.09 of the run length (median); on all other groups the channel remains schedulable to the end, with the quota level set by the survival feedback.

TABLE S7b: Survival rate of injected reflection candidates per group (per injection event, pooled over all runs of the group).

Group	Events	Median survival	IQR
MMF	2353	0.167	0.133
IDMP	2298	0.078	0.075
Local-front	3007	0.122	0.167
Polygon	758	0.033	0.100
Mode-count	1143	0.083	0.068
HDMMF	653	0.000	0.017

This gate pins the property used by the architecture ablation of Section S5: the comparison against DS-MMEA-1F isolates the second family and nothing else.

- 2) **Backbone reduction.** With the mode-relative selection layer and the manifold channel disabled, the code must reproduce a reference implementation of the backbone bit for bit on the same fixed set. This gate guarantees that the reflection and budget machinery of the discovery channel is exercised exactly as it is in the reference implementation.

Together the two gates chain: the shipped algorithm, run at family count 1 with its upper layers disabled, is bit-identical to the reference implementation, so no silent drift can enter through subsequent code changes.

D. Floating-point drift across execution contexts

Bit-level reproducibility holds within one execution context: the same seed, the same code, and the same context — all runs of the archived comparisons executed inside the parallel worker pool — reproduce identical result files. Across contexts the reproduction is numeric but not bit-level: a run executed serially and the same seed executed in a parallel worker can diverge because BLAS routines may reorder floating-point summations, and a single reordered sum propagates through the stochastic stream. All paired statistics in this paper therefore compare runs produced in the same context, with the documented exception of the re-run manifold configuration (Section S5, row 2), whose verdicts nevertheless reproduce the earlier record exactly. The per-run archives — one file per run, holding the final population, archive checkpoints at every 20% of the budget, and the per-generation mechanism logs — are the single source from which every reported statistic and every figure of this paper is computed. Re-running the pipeline in a different context may shift individual run trajectories but reproduces the aggregates to within the run-to-run variability reported in the paired tests.

S7. SEARCH-BEHAVIOR ANALYSIS: CHANNEL ACTIVITY AND FAMILY DIVISION

This section documents the search-behavior analysis behind Section VII of the main text, computed from the archived results of the same runs (per-generation mechanism logs and budget checkpoints); no additional evaluations were performed.

Channel activity over the benchmark: Table S7a summarizes the manifold channel per group: the share of generations in which it produces offspring, its share of the offspring pool, and its share in the early/middle/late thirds of the run. The channel is a genuine producer on five groups (85–92% of generations active, 13–30% of offspring, peaking in the middle third); on the HDMMF group it never matures a single model (both families fragment into two-member clusters that never reach the maturity threshold of three generations of age and eight members), and its activity ends within the first tenth of the run. Table S7b reports the survival rate of injected reflection candidates per group: single-digit to teen percentages, decreasing toward the chaotic and high-dimensional groups, and effectively zero on the HDMMF group. The survival feedback that schedules the injections (main text, Section V-E) therefore throttles them precisely where they do not pay.

The manifold channel in operation: A snapshot panel (six problems, one run each, five time points, both families) records the parent population and the mode tables of both families together with the manifold candidates they produced. Three regimes appear. Where the channel learns the shape of a mode, its samples land closer than the population: on `Polygon_M3_D4` the tight family holds four mature models (one per true Pareto set) whose samples sit at a median distance of 0.0099 against 0.0196 for the population, and on `IDMPT1_NP8` the tight family grows from one to seven mature models (against eight true modes) with samples at the population’s own final distance of 0.0002. Where the models go bad, the feedback withdraws the

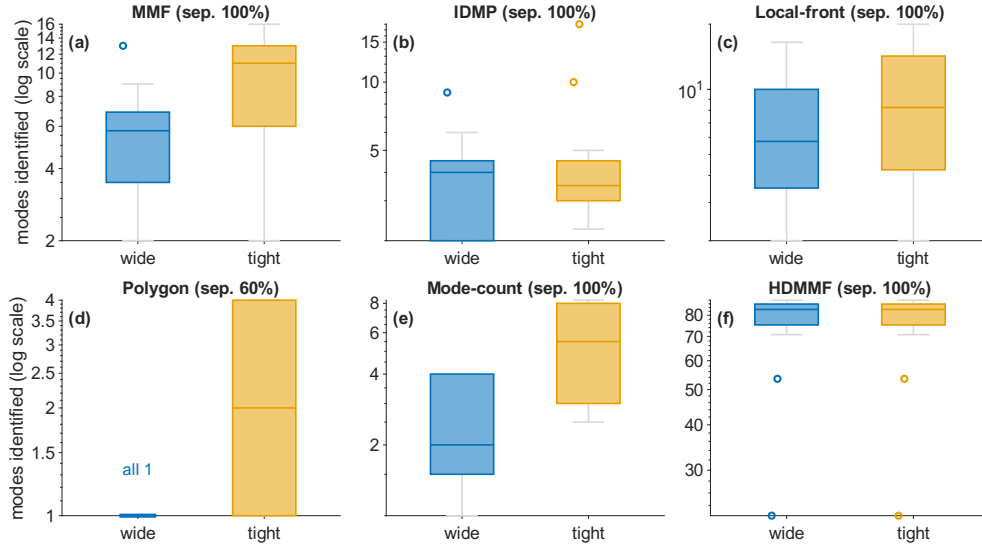


Fig. S7-1: Division of labor between the families across the benchmark, from the per-generation mechanism logs of the archived runs. Each panel shows the late-stage (last third of generations) median mode count per family and problem (log scale; box = interquartile range over the problems of the group, dots = single problems). Wide family in blue, tight family in orange; the percentage in each panel title is the separation rate (share of runs in which the tight family tracks at least half the true mode count). On the HDMMF group both families fragment into many micro-clusters, so the threshold is crossed without structure; on the Polygon group only 60% of runs separate—the behavioral boundary of the tight scale (main text, Section VIII).

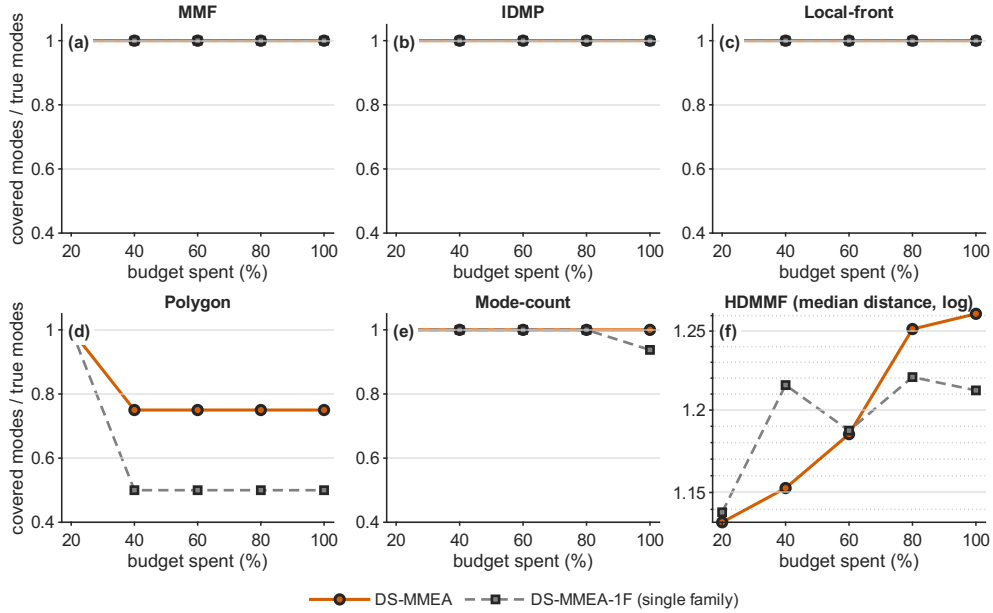


Fig. S7-2: Delivered-set coverage (covered true modes over the true mode count; group medians over problems and runs) at the five 20% budget checkpoints: DS-MMEA (solid) against the single-family counterpart DS-MMEA-1F (dashed). Modes are found early everywhere; the Polygon group sheds coverage mid-run under both algorithms (chaotic trajectories) and DS-MMEA keeps more of it, and the mode-count group shows the single-family counterpart losing one mode at the end. The HDMMF panel reports the median box-normalized distance to the true set (log scale), because the membership threshold degenerates at that dimensionality.

quota: on MMF4 the samples of the wide family sit an order of magnitude farther than its population (0.0645 against 0.0060), and the candidate count decays from 40 to 15; on MMF1 the channel produces 19 candidates at mid-run and none at the end. And where the search itself does not converge, the channel never starts: on NMFF1 both families sit at a distance of 1.66 from the true set, no mode ever matures, and no candidate is ever produced, the degenerate regime that Table S7a shows at the scale of the whole HDMMF group.

Reproducibility: All statistics in this section are computed by a single script from the archived per-run archives (mechanism logs, checkpoints, and the snapshot panel); the aggregate bundle and the analysis scripts are part of the code repository (Section S6).

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